Forecasting Avoidable Deaths from ED Admission Delays

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1. Introduction and Model Choice

This report describes a horizon-weighted ensemble of two sparse multitask Gaussian Process models submitted to the SPHERE-PPL NHS EAD Forecasting Competition. The task is to predict daily estimated avoidable deaths from ED admission delays across the Bristol NHS system, ten days ahead, over 173 sliding windows.

The submitted model — **gp_ensemble** — combines two sparse RQ-iso multitask GPs: *gp_calendar* (standard calendar lags) and *gp_informed_lags* (consecutive short lags chosen to respect the autocorelation structure of the outcome variable). Ensemble weights are derived separately for forecast horizons 1–5 and 6–10 by inverse-MSE weighting on a held-in training evaluation.

The architecture was selected after evaluating GP-family and tree-based candidates on a held-out hard-week / average-difficulty-week split drawn from 2022–2024 development data; GPs were favoured for their robustness under distribution shift on the hardest weeks. Full architectural sweep details are documented in the development repository.

The final submitted forecast augments this ensemble with a seasonal-naïve **persistence** component and a non-negativity clamp:

$$\text{final} = 0.75\,\hat{y}_{\text{GP}} + 0.25\,\hat{y}_{\text{persist}}, \qquad \text{then} \quad \max(\cdot,\, 0).$$

The persistence term is weekday-aligned — it copies the outcome from one week earlier for horizons 1–3 and two weeks earlier for horizons 4–10 — using only values at $D-4$ or older, so it respects the same availability cutoff as the GP features. It corrects a mild under-dispersion of the GP posterior mean in elevated winters by reinjecting the recent level; because the GP and persistence errors are largely decorrelated, the blend also reduces variance. The weight 0.25 is deliberately conservative — enough recent level to correct the winter under-dispersion while keeping the GP dominant, which limits the downside in weeks where the recent level is a poor guide (such as sharp declines).

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2. Data Pipeline

The development dataset spans 16 March 2023 to 30 September 2025 (930 daily observations). The outcome variable is daily estimated avoidable deaths with no missing values. Of 220 candidate sensor variables, the three most predictive DTA columns (Bristol Royal Infirmary, Weston, North Bristol

NHS Trust) were selected based on Spearman correlation with the outcome across all candidate sensors. Addition of further sensors led to performance deterioration.

Sensor readings arriving after noon on day D are attributed to $D+1$ to respect the competition's midday cutoff. A three-day outcome reporting lag means only y_{D-4} and earlier are used as input features (one day more conservative than the competition rule allows). Rolling means and outcome lag features are anchored at $D-4$ consistently across both models. Missing sensor values within the training set are forward-filled, backward-filled, then zero-filled.

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3. Model Structure

Both component models share the `MultitaskSparseRqIsoGP` architecture: ten independent sparse GPs (one per forecast horizon) implemented via `IndependentMultitaskVariationalStrategy` with a `ScaleKernel(RQKernel)` covariance and Cholesky variational distribution. The mean function is a per-task `ConstantMean`, leaving the RQ kernel to model all structure in the data. Training maximises the ELBO via Adam with cosine-annealed learning rate and mini-batch data loading. Models are trained once on the full development dataset; prediction over 173 assessment windows runs in seconds per window.

The two models differ only in their feature sets:

	gp_calendar	**gp_informed_lags**
Outcome lags	4, 7, 14, 28 days	4–7, 13–14, 20–21 days + $\Delta_{4,7}$
Sensor lags	0, 3, 7, 28 days (offset by outcome lag)	0–2, 6–7, 13–14, 20–21 days (raw)
Rolling means	7, 14, 28-day windows at $D-4$	7, 14, 28-day windows at $D-4$
lr / inducing / n_iter	0.054 / 100 / 200	0.1 / 100 / 200

Hyperparameters and the choice of `ConstantMean` over a richer mean function were selected on a held-out split of the development period, documented in the development repository.

Persistence blend. Let $\text{lag}(h) = 7$ for $h \leq 3$ and 14 otherwise. The persistence forecast is $\hat{y}_{\text{persist}}[D, h] = y_{D+h-\text{lag}(h)}$, whose freshest reach is y_{D-4} (attained at $h = 3$ and $h = 10$), within the availability cutoff. The shipped forecast is $0.75\,\hat{y}_{\text{GP}} + 0.25\,\hat{y}_{\text{persist}}$, clamped to be non-negative. The clamp is no-regret under MSE — the outcome is non-negative, so flooring a negative posterior mean to zero can only reduce squared error — and in practice it almost never activates.

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4. Evaluation

The table and trajectory plot below are in-sample over the last 30 training windows — the model was trained on these windows, so MSE values are optimistic relative to the out-of-sample assessment period the contest will score on. They are included as a sanity check that the ensemble fits the training distribution and that horizon-specific weighting yields a coherent improvement over either component model. The bolded row is the **shipped** model (GP ensemble plus the 0.25 persistence blend, clamped); the unbolded `gp_ensemble` row is the pre-blend GP for comparison. Out-of-sample generalisation was validated separately during development on a held-out 2025-04-01 – 2025-09-30 split, with results documented in the development repository.

Model	MSE 1–5d	MSE 6–10d
`gp_calendar`	0.050127	0.033071
`gp_informed_lags`	0.046068	0.033528
`gp_ensemble` (pre-blend)	0.047200	0.032685
SHIPPED: gp_ensemble + 0.25 persistence	**0.051114**	**0.032431**
Predict-last	0.100243	0.072142
Predict-mean	0.057300	0.060574

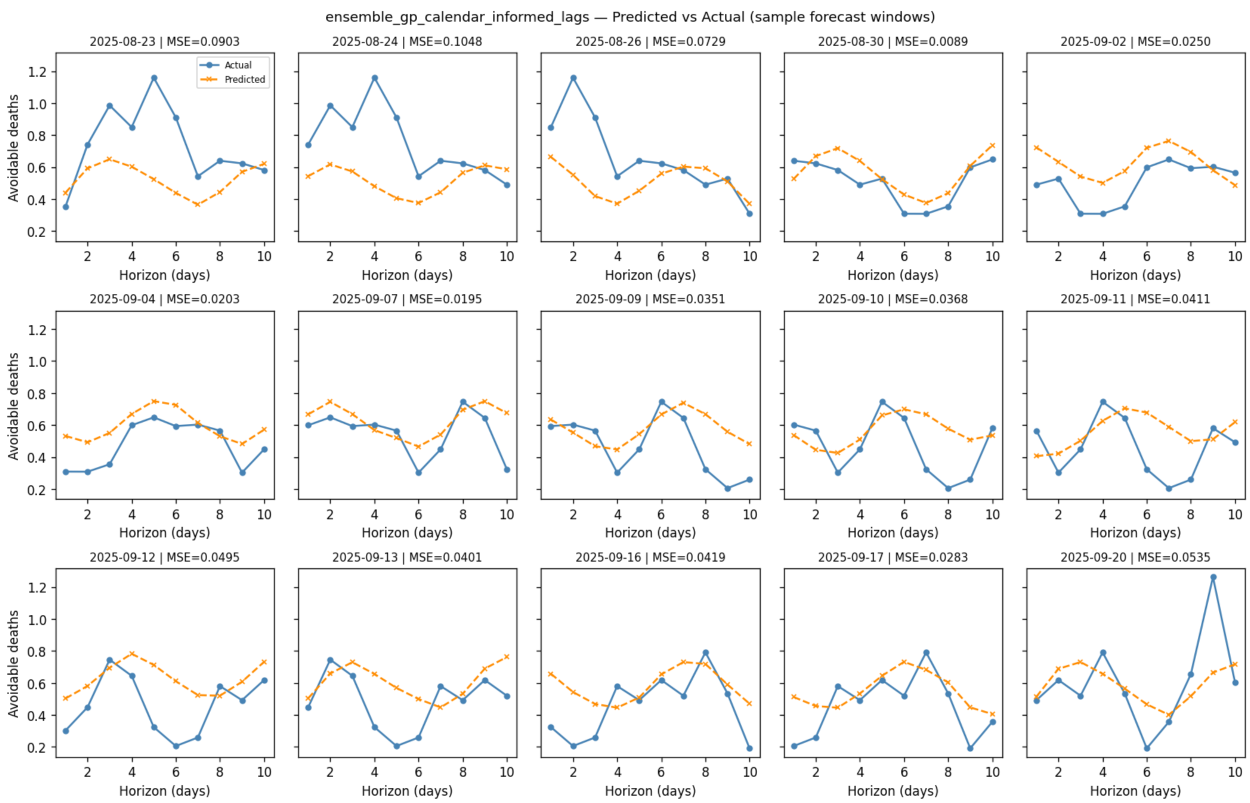

Figure 1: Actual (blue) vs predicted (orange dashed) trajectories for 15 sampled training windows over the 10-day forecast horizon.

5. Runtime

End-to-end pipeline timing for the run that produced this report. Captured on CPU.

Step	Time (s)
`step_2_load_data`	27.0
`step_3_define_windows`	0.0
`step_4_build_features`	0.3
`step_5_train_models`	135.1
`step_6_predict`	0.1
`step_7_write_csv`	0.0
`step_8_mse_summary`	0.0
`pipeline_total`	165.6
(report compile not included)	